

LibreOffice 26.2 Help

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Statistical Functions Part One

COUNTIFS

Returns the count of cells that meet criteria in multiple ranges.

B

Returns the probability of a sample with binomial distribution.

Syntax

$B(\text{Trials}; SP; T1 \text{ [}; T2])$

Trials is the number of independent trials.

SP is the probability of success on each trial.

T1 defines the lower limit for the number of trials.

T2 (optional) defines the upper limit for the number of trials.

Example

What is the probability with ten throws of the dice, that a six will come up exactly twice? The probability of a six (or any other number) is 1/6. The following formula combines these factors:

$=B(10; 1/6; 2)$ returns a probability of 29%.

BETA.DIST

Returns the beta function.



This function is available since LibreOffice 4.2.

Syntax

`BETA.DIST(Number; Alpha; Beta; Cumulative [; Start [; End]])`

Number (required) is the value between **Start** and **End** at which to evaluate the function.

Alpha (required) is a parameter to the distribution.

Beta (required) is a parameter to the distribution.

Cumulative (required) can be 0 or False to calculate the probability density function. It can be any other value or True to calculate the cumulative distribution function.

Start (optional) is the lower bound for **Number**.

End (optional) is the upper bound for **Number**.

In the LibreOffice Calc functions, parameters marked as "optional" can be left out only when no parameter follows. For example, in a function with four parameters, where the last two parameters are marked as "optional", you can leave out parameter 4 or parameters 3 and 4, but you cannot leave out parameter 3 alone.

Example

`=BETA.DIST(2;8;10;1;1;3)` returns the value 0.6854706

`=BETA.DIST(2;8;10;0;1;3)` returns the value 1.4837646

Technical information

This function is **NOT** part of the **Open Document Format for Office Applications (OpenDocument) Version 1.3. Part 4: Recalculated Formula (OpenFormula) Format** standard. The name space is

COM.MICROSOFT.BETA.DIST

BETADIST

Returns the beta function.

Syntax

 **BETADIST**(Number; Alpha; Beta [; Start [; End [; Cumulative]]])

Number is the value between **Start** and **End** at which to evaluate the function.

Alpha is a parameter to the distribution.

Beta is a parameter to the distribution.

Start (optional) is the lower bound for **Number**.

End (optional) is the upper bound for **Number**.

Cumulative (optional) can be 0 or False to calculate the probability density function. It can be any other value or True or omitted to calculate the cumulative distribution function.

In the LibreOffice Calc functions, parameters marked as "optional" can be left out only when no parameter follows. For example, in a function with four parameters, where the last two parameters are marked as "optional", you can leave out parameter 4 or parameters 3 and 4, but you cannot leave out parameter 3 alone.

Example

=BETADIST(0.75;3;4) returns the value 0.96.

BETA.INV

Returns the inverse of the cumulative Beta probability density function.

Syntax

BETA.INV(Number; Alpha; Beta [; Start [; End]])

Number is the probability associated with the Beta distribution for the given arguments **Alpha** and **Beta**.

Alpha is a strictly positive parameter of the Beta distribution.

Beta is a strictly positive parameter of the Beta distribution.

Start (optional) is the lower bound of the output range of the function. If omitted, the default value is 0.

End (optional) is the upper bound of the output range of the function. If omitted, the default value is 1.

In the LibreOffice Calc functions, parameters marked as "optional" can be left out only when no parameter follows. For example, in a function with four parameters, where the

 Last two parameters are marked as "optional", you can leave out parameter 4 or parameters 3 and 4, but you cannot leave out parameter 3 alone.

Example

`=BETA.INV(0.5;5;10)` returns the value 0.3257511553.

[BETA.INV Wiki page](#)

Technical information



This function is available since LibreOffice 4.2.

This function is **NOT** part of the **Open Document Format for Office Applications (OpenDocument) Version 1.3. Part 4: Recalculated Formula (OpenFormula) Format** standard. The name space is

COM.MICROSOFT.BETA.INV

BETAINV

Returns the inverse of the cumulative Beta probability density function.

Syntax

`BETAINV(Number; Alpha; Beta [; Start [; End]])`

Number is the probability associated with the Beta distribution for the given arguments **Alpha** and **Beta**.

Alpha is a strictly positive parameter of the Beta distribution.

Beta is a strictly positive parameter of the Beta distribution.

Start (optional) is the lower bound of the output range of the function. If omitted, the default value is 0.

End (optional) is the upper bound of the output range of the function. If omitted, the default value is 1.

In the LibreOffice Calc functions, parameters marked as "optional" can be left out only when no parameter follows. For example, in a function with four parameters, where the last two parameters are marked as "optional", you can leave out parameter 4 or parameters 3 and 4, but you cannot leave out parameter 3 alone.

Example

`=BETAINV(0.5;5;10)` returns the value 0.3257511553.

[BETAINV Wiki page](#)

BINOM.DIST

Returns the individual term binomial distribution probability.

Syntax

`BINOM.DIST(X; Trials; SP; C)`

X is the number of successes in a set of trials.

Trials is the number of independent trials.

SP is the probability of success on each trial.

C = 0 calculates the probability of a single event and **C** = 1 calculates the cumulative probability.

Example

`=BINOM.DIST(A1;12;0.5;0)` shows (if the values 0 to 12 are entered in A1) the probabilities for 12 flips of a coin that **Heads** will come up exactly the number of times entered in A1.

`=BINOM.DIST(A1;12;0.5;1)` shows the cumulative probabilities for the same series. For example, if A1 = 4, the cumulative probability of the series is 0, 1, 2, 3 or 4 times **Heads** (non-exclusive OR).

Technical information



This function is available since LibreOffice 4.2.

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COM.MICROSOFT.BINOM.DIST

BINOMDIST

 Returns the individual term binomial distribution probability.

Syntax

`BINOMDIST(X; Trials; SP; C)`

X is the number of successes in a set of trials.

Trials is the number of independent trials.

SP is the probability of success on each trial.

C = 0 calculates the probability of a single event and **C** = 1 calculates the cumulative probability.

Example

`=BINOMDIST(A1;12;0.5;0)` shows (if the values 0 to 12 are entered in A1) the probabilities for 12 flips of a coin that **Heads** will come up exactly the number of times entered in A1.

`=BINOMDIST(A1;12;0.5;1)` shows the cumulative probabilities for the same series. For example, if A1 = 4, the cumulative probability of the series is 0, 1, 2, 3 or 4 times **Heads** (non-exclusive OR).

BINOM.INV

Returns the smallest value for which the cumulative binomial distribution is greater than or equal to a criterion value.

Syntax

`BINOM.INV(Trials; SP; Alpha)`

Trials The total number of trials.

SP is the probability of success on each trial.

Alpha The border probability that is attained or exceeded.

Example

`=BINOM.INV(8;0.6;0.9)` returns 7, the smallest value for which the cumulative binomial distribution is greater than or equal to a criterion value.

Technical information



This function is available since LibreOffice 4.2.

This function is **NOT** part of the **Open Document Format for Office Applications (OpenDocument) Version 1.3. Part 4: Recalculated Formula (OpenFormula) Format** standard. The name space is

COM.MICROSOFT.BINOM.INV

CHIDIST

Returns the probability value from the indicated Chi square that a hypothesis is confirmed. CHIDIST compares the Chi square value to be given for a random sample that is calculated from the sum of (observed value-expected value)²/expected value for all values with the theoretical Chi square distribution and determines from this the probability of error for the hypothesis to be tested.

The probability determined by CHIDIST can also be determined by CHITEST.

Syntax

CHIDIST(Number; DegreesFreedom)

Number is the chi-square value of the random sample used to determine the error probability.

DegreesFreedom are the degrees of freedom of the experiment.

Example

=CHIDIST(13.27; 5) equals 0.02.

If the Chi square value of the random sample is 13.27 and if the experiment has 5 degrees of freedom, then the hypothesis is assured with a probability of error of 2%.

CHIINV

Returns the inverse of the one-tailed probability of the chi-squared distribution.

Syntax

CHIINV(Number; DegreesFreedom)

Number is the value of the error probability.

 **DegreesFreedom** is the degrees of freedom of the experiment.

Example

A die is thrown 1020 times. The numbers on the die 1 through 6 come up 195, 151, 148, 189, 183 and 154 times (observation values). The hypothesis that the die is not fixed is to be tested.

The Chi square distribution of the random sample is determined by the formula given above. Since the expected value for a given number on the die for n throws is n times 1/6, thus $1020/6 = 170$, the formula returns a Chi square value of 13.27.

If the (observed) Chi square is greater than or equal to the (theoretical) Chi square **CHIINV**, the hypothesis will be discarded, since the deviation between theory and experiment is too great. If the observed Chi square is less than **CHIINV**, the hypothesis is confirmed with the indicated probability of error.

`=CHIINV(0.05;5)` returns 11.07.

`=CHIINV(0.02;5)` returns 13.39.

If the probability of error is 5%, the die is not true. If the probability of error is 2%, there is no reason to believe it is fixed.

CHISQ.DIST

Returns the probability density function or the cumulative distribution function for the chi-square distribution.

Syntax

CHISQ.DIST(Number; DegreesFreedom; Cumulative)

Number is the chi-square value of the random sample used to determine the error probability.

DegreesFreedom are the degrees of freedom of the experiment.

Cumulative can be 0 or False to calculate the probability density function. It can be any other value or True to calculate the cumulative distribution function.

Example

`=CHISQ.DIST(3; 2; 0)` equals 0.1115650801, the probability density function with 2 degrees of freedom, at $x = 3$.

=CHISQ.DIST(3; 2; 1) equals 0.7768698399, the cumulative chi-square distribution with 2 degrees of freedom, at the value $x = 3$.

Technical information



This function is available since LibreOffice 4.2.

This function is **NOT** part of the **Open Document Format for Office Applications (OpenDocument) Version 1.3. Part 4: Recalculated Formula (OpenFormula) Format** standard. The name space is

COM.MICROSOFT.CHISQ.DIST

CHISQDIST

Returns the value of the probability density function or the cumulative distribution function for the chi-square distribution.

Syntax

CHISQDIST(Number; Degrees Of Freedom [; Cumulative])

Number is the number for which the function is to be calculated.

Degrees Of Freedom is the degrees of freedom for the chi-square function.

Cumulative (optional): 0 or False calculates the probability density function. Other values or True or omitted calculates the cumulative distribution function.

CHISQ.DIST.RT

Returns the probability value from the indicated Chi square that a hypothesis is confirmed. CHISQ.DIST.RT compares the Chi square value to be given for a random sample that is calculated from the sum of $(\text{observed value} - \text{expected value})^2 / \text{expected value}$ for all values with the theoretical Chi square distribution and determines from this the probability of error for the hypothesis to be tested.

The probability determined by CHISQ.DIST.RT can also be determined by CHITEST.

Syntax

CHISQ.DIST.RT(Number; DegreesFreedom)

Number is the chi-square value of the random sample used to determine the error probability.

DegreesFreedom are the degrees of freedom of the experiment.

Example

`=CHISQ.DIST.RT(13.27; 5)` equals 0.0209757694.

If the Chi square value of the random sample is 13.27 and if the experiment has 5 degrees of freedom, then the hypothesis is assured with a probability of error of 2%.

Technical information



This function is available since LibreOffice 4.2.

This function is **NOT** part of the **Open Document Format for Office Applications (OpenDocument) Version 1.3. Part 4: Recalculated Formula (OpenFormula) Format** standard. The name space is

COM.MICROSOFT.CHISQ.DIST.RT

CHISQ.INV

Returns the inverse of the left-tailed probability of the chi-square distribution.

Syntax

`CHISQ.INV(Probability; DegreesFreedom)`

Probability is the probability value for which the inverse of the chi-square distribution is to be calculated.

Degrees Of Freedom is the degrees of freedom for the chi-square function.

Example

`=CHISQ.INV(0,5;1)` returns 0.4549364231.

Technical information



This function is available since LibreOffice 4.2.



This function is **NOT** part of the **Open Document Format for Office Applications (OpenDocument) Version 1.3. Part 4: Recalculated Formula (OpenFormula) Format** standard. The name space is

COM.MICROSOFT.CHISQ.INV

CHISQINV

Returns the inverse of CHISQDIST.

Syntax

CHISQINV(Probability; Degrees of Freedom)

Probability is the probability value for which the inverse of the chi-square distribution is to be calculated.

Degrees Of Freedom is the degrees of freedom for the chi-square function.

CHISQ.INV.RT

Returns the inverse of the one-tailed probability of the chi-squared distribution.

Syntax

CHISQ.INV.RT(Number; DegreesFreedom)

Number is the value of the error probability.

DegreesFreedom is the degrees of freedom of the experiment.

Example

A die is thrown 1020 times. The numbers on the die 1 through 6 come up 195, 151, 148, 189, 183 and 154 times (observation values). The hypothesis that the die is not fixed is to be tested.

The Chi square distribution of the random sample is determined by the formula given above. Since the expected value for a given number on the die for n throws is n times 1/6, thus $1020/6 = 170$, the formula returns a Chi square value of 13.27.

If the (observed) Chi square is greater than or equal to the (theoretical) Chi square CHIIINV, the hypothesis will be discarded, since the deviation between theory and

 experiment is too great. If the observed Chi square is less than CHINVT, the hypothesis is confirmed with the indicated probability of error.

`=CHISQ.INV.RT(0.05;5)` returns 11.0704976935.

`=CHISQ.INV.RT(0.02;5)` returns 13.388222599.

If the probability of error is 5%, the die is not true. If the probability of error is 2%, there is no reason to believe it is fixed.

Technical information



This function is available since LibreOffice 4.2.

This function is **NOT** part of the **Open Document Format for Office Applications (OpenDocument) Version 1.3. Part 4: Recalculated Formula (OpenFormula) Format** standard. The name space is

COM.MICROSOFT.CHISQ.INV.RT

CHISQ.TEST

Returns the probability of a deviance from a random distribution of two test series based on the chi-squared test for independence. CHISQ.TEST returns the chi-squared distribution of the data.

The probability determined by CHISQ.TEST can also be determined with CHISQ.DIST, in which case the Chi square of the random sample must then be passed as a parameter instead of the data row.

Syntax

`CHISQ.TEST(DataB; DataE)`

DataB is the array of the observations.

DataE is the range of the expected values.

Example

	Data_B (observed)	Data_E (expected)
1	195	170



2	151	170
3	148	170
4	189	170
5	183	170
6	154	170

`=CHISQ.TEST(A1:A6;B1:B6)` equals 0.0209708029. This is the probability which suffices the observed data of the theoretical Chi-square distribution.

Technical information



This function is available since LibreOffice 4.2.

This function is **NOT** part of the **Open Document Format for Office Applications (OpenDocument) Version 1.3. Part 4: Recalculated Formula (OpenFormula) Format** standard. The name space is

COM.MICROSOFT.CHISQ.TEST

CHITEST

Returns the probability of a deviance from a random distribution of two test series based on the chi-squared test for independence. CHITEST returns the chi-squared distribution of the data.

The probability determined by CHITEST can also be determined with CHIDIST, in which case the Chi square of the random sample must then be passed as a parameter instead of the data row.

Syntax

`CHITEST(DataB; DataE)`

DataB is the array of the observations.

DataE is the range of the expected values.

Example

	Data_B (observed)	Data_E (expected)
1	195	170
2	151	170
3	148	170
4	189	170
5	183	170
6	154	170

=CHITEST(A1:A6;B1:B6) equals 0.02. This is the probability which suffices the observed data of the theoretical Chi-square distribution.

COUNT

Counts how many numbers are in the list of arguments. Text entries are ignored.

Syntax

COUNT(Argument 1 [; Argument 2 [; ... [; Argument 255]]])

Argument 1; Argument 2; ... ; Argument 255 are numbers, text, references to cells or to cell ranges of numbers or text.



This function ignores any text or empty cell within a data range. If you suspect wrong results from this function, look for text in the data ranges. To highlight text contents in a data range, use the [value highlighting](#) feature.

Example

The entries 2, 4, 6 and "eight" in the Value 1-4 fields are to be counted.

=COUNT(2;4;6;"eight") = 3. The count of numbers is therefore 3.

COUNTA

Counts how many values are in the list of arguments. Text entries are also counted, even when they contain an empty string of length 0. If an argument is an array or reference, empty cells within the array or reference are ignored.

Syntax

COUNTA(Argument 1 [; Argument 2 [; ... [; Argument 255]]])

Argument 1; Argument 2; ... ; Argument 255 are numbers, text, references to cells or to cell ranges of numbers or text.

Example

The entries 2, 4, 6 and "eight" in the Value 1-4 fields are to be counted.

=COUNTA(2;4;6;"eight") = 4. The count of values is therefore 4.

COUNTBLANK

Returns the number of empty cells.

Syntax

COUNTBLANK(Range)

Returns the number of empty cells in the cell range **Range**.

Example

=COUNTBLANK(A1:B2) returns 4 if cells A1, A2, B1, and B2 are all empty.

COUNTIF

Returns the number of cells that meet with certain criteria within a cell range.

Syntax

COUNTIF(Range; Criterion)

Range is the range to which the criteria are to be applied.

Criterion1: A criterion is a text expression computed from a single cell Reference, Formula, Number or Text. It is used in comparisons with cell contents.

 A reference to an empty cell is interpreted as the numeric value 0.

A matching expression can be:

- A number or logical value. A matching cell content equals the number or logical value.
- A value beginning with a comparator (<, <=, =, >, >=, <>).

For =, if the value is empty it matches empty cells.

For <>, if the value is empty it matches non-empty cells.

For <>, if the value is not empty it matches any cell content except the value, including empty cells.

Note: "=0" does not match empty cells.

For = and <>, if the value is not empty and can not be interpreted as a number type or one of its subtypes and the property [Search criteria = and <> must apply to whole cells](#) is checked, comparison is against the entire cell contents, if unchecked, comparison is against any subpart of the field that matches the criteria. For = and <>, if the value is not empty and can not be interpreted as a Number type or one of its subtypes applies.

- Other Text value. If the property [Search criteria = and <> must apply to whole cells](#) is true, the comparison is against the entire cell contents, if false, comparison is against any subpart of the field that matches the criteria. The expression can contain text, numbers, regular expressions or wildcards ([if enabled in calculation options](#)).



The search supports wildcards or [regular expressions](#). With regular expressions enabled, you can enter "all.*", for example to find the first location of "all" followed by any characters. If you want to search for a text that is also a regular expression, you must either precede every regular expression metacharacter or operator with a "\" character, or enclose the text into \Q...\E. You can switch the automatic evaluation of wildcards or regular expression on and off in [Tools - Options - LibreOffice Calc - Calculate](#).

When using functions where one or more arguments are search criteria strings that represents a regular expression, the first attempt is to convert the string criteria to numbers. For example, ".0" will convert to 0.0 and so on. If successful,



the match will not be a regular expression match but a numeric match. However, when switching to a locale where the decimal separator is not the dot makes the regular expression conversion work. To force the evaluation of the regular expression instead of a numeric expression, use some expression that can not be misread as numeric, such as "[0]" or ".\0" or "(?i).0".

Example

A1:A10 is a cell range containing the numbers 2000 to 2009. Cell B1 contains the number 2006. In cell B2, you enter a formula:

=COUNTIF(A1:A10;2006) - this returns 1.

=COUNTIF(A1:A10;B1) - this returns 1.

=COUNTIF(A1:A10;">=2006") - this returns 4.

=COUNTIF(A1:A10;"<"&B1) - when B1 contains 2006, this returns 6.

=COUNTIF(A1:A10;C2) where cell C2 contains the text >2006 counts the number of cells in the range A1:A10 which are >2006.

To count only negative numbers: =COUNTIF(A1:A10;"<0")

EXPON.DIST

Returns the exponential distribution.

Syntax

EXPON.DIST(Number; Lambda; C)

Number is the value of the function.

Lambda is the parameter value.

C is a logical value that determines the form of the function. **C = 0** calculates the density function, and **C = 1** calculates the distribution.

Example

=EXPON.DIST(3;0.5;1) returns 0.7768698399.

Technical information



This function is available since LibreOffice 4.2.

This function is **NOT** part of the **Open Document Format for Office Applications (OpenDocument) Version 1.3. Part 4: Recalculated Formula (OpenFormula) Format** standard. The name space is

COM.MICROSOFT.EXPON.DIST

EXPONDIST

Returns the exponential distribution.

Syntax

EXPONDIST(Number; Lambda; C)

Number is the value of the function.

Lambda is the parameter value.

C is a logical value that determines the form of the function. **C = 0** calculates the density function, and **C = 1** calculates the distribution.

Example

=EXPONDIST(3;0.5;1) returns 0.78.

INTERCEPT

Calculates the point at which a line will intersect the y-values by using known x-values and y-values.

Syntax

INTERCEPT(DataY; DataX)

DataY is the dependent set of observations or data.

DataX is the independent set of observations or data.

Names, arrays or references containing numbers must be used here. Numbers can also be entered directly.

Example

To calculate the intercept, use cells D3:D9 as the y value and C3:C9 as the x value from the example spreadsheet. Input will be as follows:

`=INTERCEPT(D3:D9;C3:C9) = 2.15.`

RSQ

Returns the square of the Pearson correlation coefficient based on the given values. RSQ (also called determination coefficient) is a measure for the accuracy of an adjustment and can be used to produce a regression analysis.

Syntax

`RSQ(DataY; DataX)`

DataY is an array or range of data points.

DataX is an array or range of data points.

Example

`=RSQ(A1:A20;B1:B20)` calculates the determination coefficient for both data sets in columns A and B.

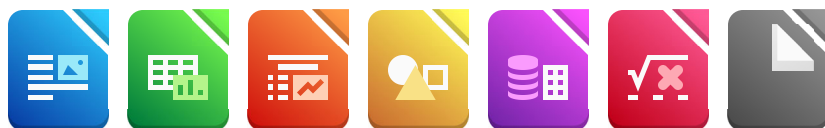
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